

Product Requirement Document (PRD)

Project: Next-Gen BPO Workforce Analytics & Productivity Optimization Platform
Target Audience: COO, Operations Leaders, and Team Leads

1. Executive Summary & Objective

This document outlines the engineering blueprint for a scalable, high-performance operations tracking platform tailored specifically for the BPO industry. The primary objective is to eliminate operational leakages (unbilled desktop down-time, compliance failures, and capacity imbalances) while preserving worker privacy and system efficiency.

2. Core Technical Architecture Blueprint

The system relies on an asymmetric hybrid architecture structured to collect endpoint data resiliently without inducing CPU throttling on target agent workstations.

- **Desktop Agent (The Probe):** A lightweight execution loop built in C#.NET or Go. Operates as a background system tray daemon tracking foreground window active focus handle changes and mouse/keyboard interrupts.
- **Data Ingestion Pipeline:** Employs an API Gateway driving telemetry directly to a column-oriented time-series store (ClickHouse / TimescaleDB) to natively manage high-throughput time-series writes.
- **Management Web Console:** Responsive multi-tenant front-end dashboard built utilizing React or Angular communicating over OAuth2 secured REST/GraphQL APIs.

3. Feature & Functional Matrix

Module	Key Functional Requirements	Technical Priority
Productivity Tracker	Tracks focus times, auto-categorizes application signatures (Productive / Waste), and counts key events without logging specific inputs.	P0 (Critical)
Resilient Cache	Local encrypted SQLite transaction block log to buffer metric state if connectivity degrades; auto-flushes on reconnect.	P0 (Critical)
Security Module	Anti-tampering thread locks. Validates system time via NTP to block clients altering clock states manually.	P1 (High)
AI Attrition Risk	Statistical drift engine monitoring work pattern deviations to signal operational burnout flags.	P2 (Medium)

4. Data Security & Regulatory Strategy

Operating tracking software inside tightly regulated environments requires heavy guardrails:

1. **Zero-Keylogging Policy:** Keystrokes are monitored strictly via rate-frequency integers. Actual character data points or secure inputs must never be captured or stored to stay compliant with PCI-DSS.
2. **PII Masking Layer:** If desktop screenshot verification features are activated by operational policy, the edge client must utilize native OCR/Computer Vision to dynamically blur customer information before data packet routing.